ATTACKTWIN AI
SECURITY DIGITAL TWIN

RUN THE DEMO MENU

01 | THE BUTTERFLY EFFECT LIVE SIMULATION

Small signal.
Big consequences.

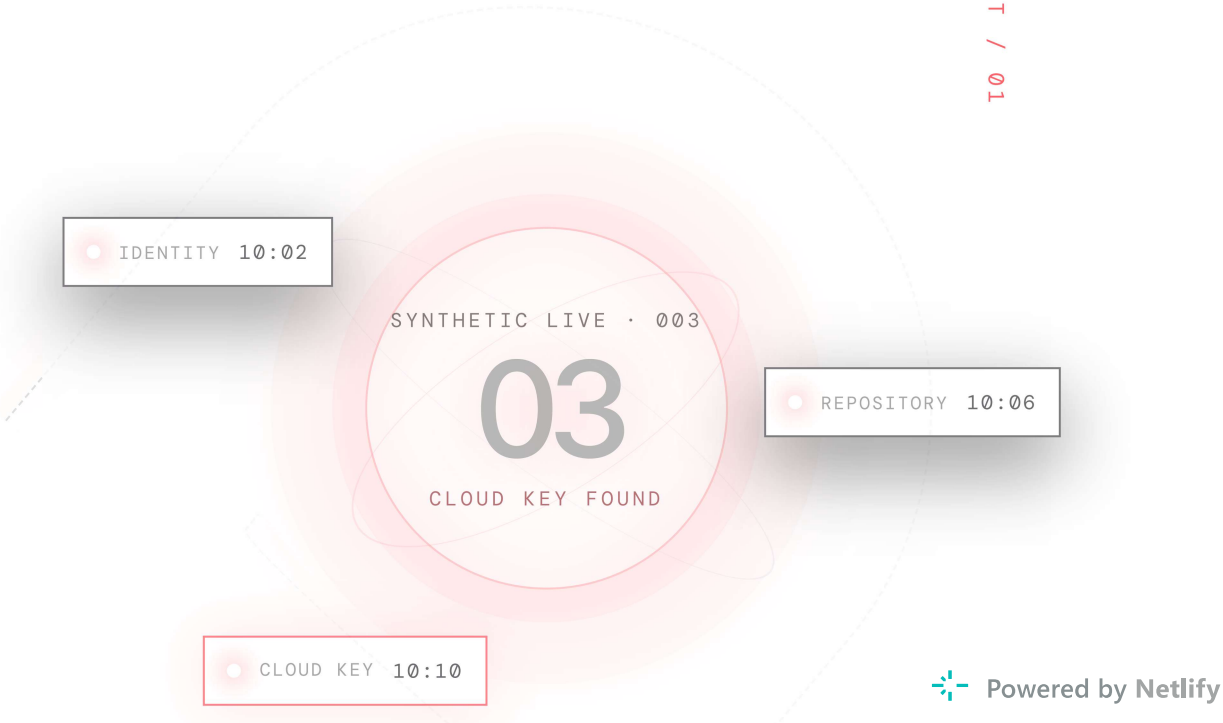
An unusual login. A forgotten cloud key. A permissive role. Individually weak signals can connect into a route to your most valuable data.

Follow the signal

What is AttackTwin? →

ONE SIMULATED SIGNAL SCROLL TO TRACE ITS IMPACT ↓

AT / 01

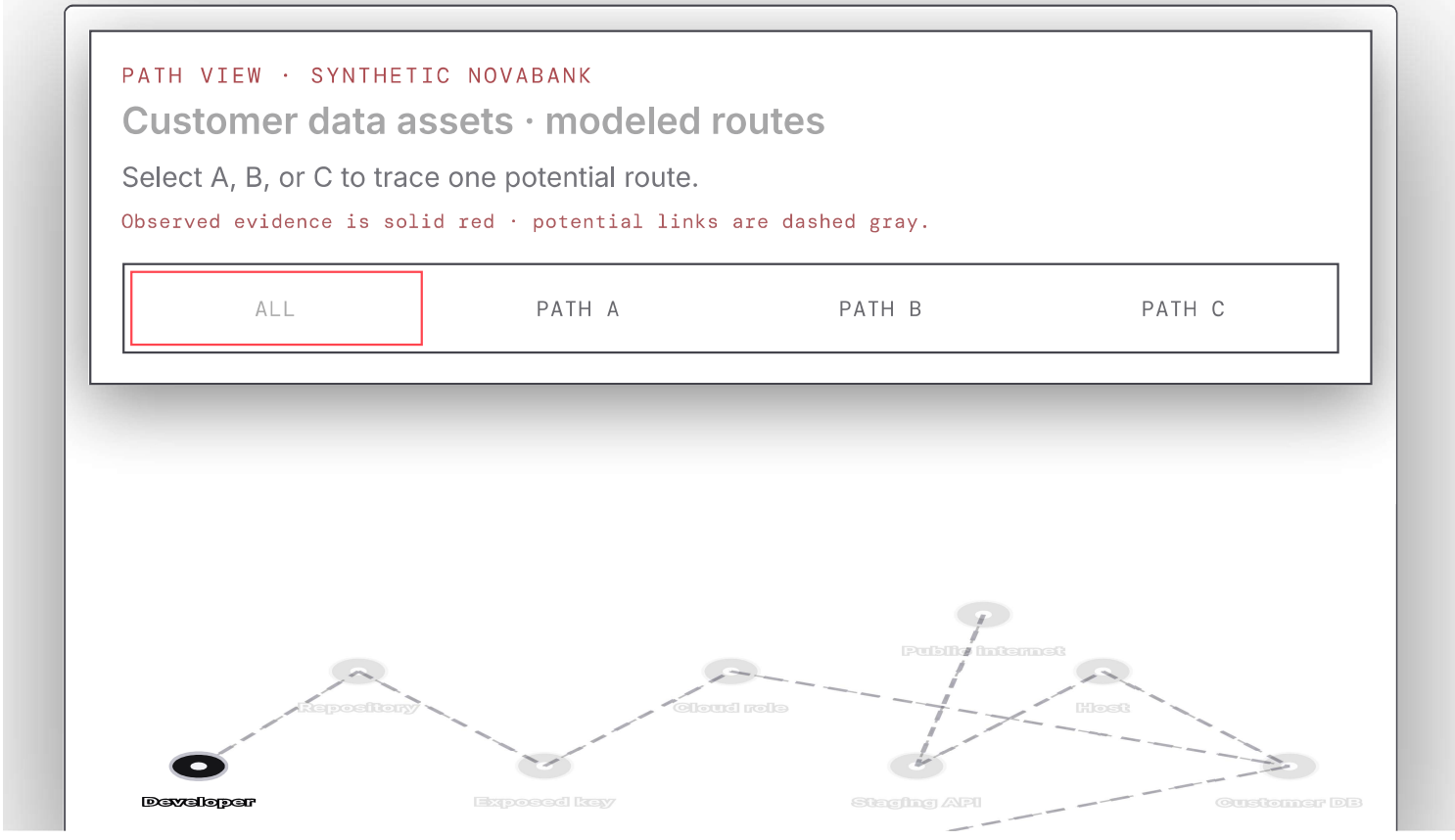


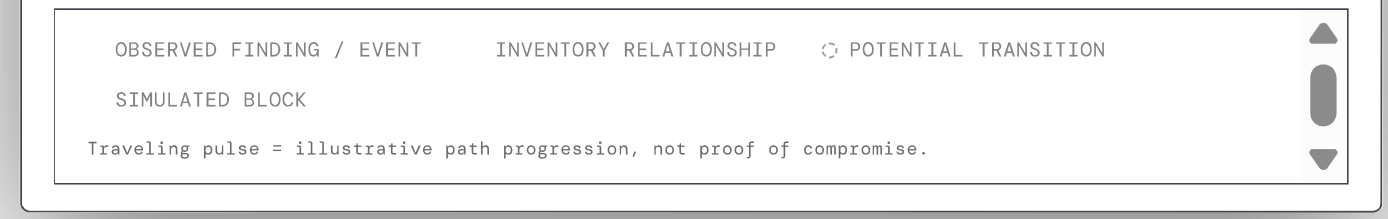
Powered by Netlify

INGESTED SIGNALS 06

Connect the alerts →

01	ID	Unusual developer login	ISOLATED
02	SC	Private repository accessed	ISOLATED
03	KY	Cloud key detected	ISOLATED
04	CL	Unexpected role assumption	ISOLATED
05	API	Exposed staging service	ISOLATED
06	DB	Abnormal database queries	ISOLATED


PATH VIEW · SYNTHETIC NOVABANK
Customer data assets · modeled routes
Select A, B, or C to trace one potential route.
Observed evidence is solid red · potential links are dashed gray.
ALL
PATH A
PATH B
PATH C
Developer
Repository
Exposed key
Cloud role
Staging API
Public Internet
Host
Customer DB



Path A	credential route	Customer DB	IDENTITY
Path B	public service route	Customer DB	APPLICATION
Path C	vendor integration	Customer DB	THIRD PARTY

Solid links are observed or inventoried. Dashed links remain potential.

- Identities
- Secrets
- Cloud resources
- Endpoints
- Applications & APIs
- Networks
- Vulnerabilities
- Third parties

COMPONENT 01 / 08

Identities

Cloud principals, role trust and associated sessions.

RELEVANT TELEMETRY

Cloud audit · IAM policy · principal inventory

EXAMPLE RELATIONSHIP

Cloud key → authenticates as → cloud principal

FEASIBILITY CHECK

Permission and reachability prerequisites

A NEW EVENT UPDATES THE GRAPH NEIGHBORHOOD IT TOUCHES.

02 Normalize & resolve →

Stable IDs · provenance

03 Temporal heterogeneous graph →

Entities · permissions · time

04 Feasibility & path compiler →

Prerequisites · bounded search

05 Priority + evidence confidence →

Potential impact · evidence · business context

06 Intervention simulation →

Cost · coverage · residuals

07 Explain & verify →

Recommendations · outcome



07 | SYNTHETIC ATTACK DEMO

Potential paths, grounded in evidence

Replay the simulated events, inspect the predicted next step, and review the conclusion.

EVENT 01 / 07

PATH A · CRITICAL · 10:02 **Exposure and permissions have strong supporting evidence.**

POTENTIAL NEXT STEP

PATH A · CRITICAL

STRONG EVIDENCE

The exposed key could authenticate as its cloud principal.

Confirm the key is active and mapped to this principal.

POTENTIAL IMPACT

Customer-data storage

RECOMMENDED NEXT STEP

Invalidate and verify

10:02 Repository secret finding
Secret scanner · OBSERVED FINDING

10:06 Cloud principal match
PENDING

10:10 Cloud role assumption recorded
PENDING

10:14 Role read permission
PENDING

10:20 Public vulnerable staging service
PENDING

10:23 Suspicious exploit attempt
PENDING

10:25 Vendor export permission
PENDING

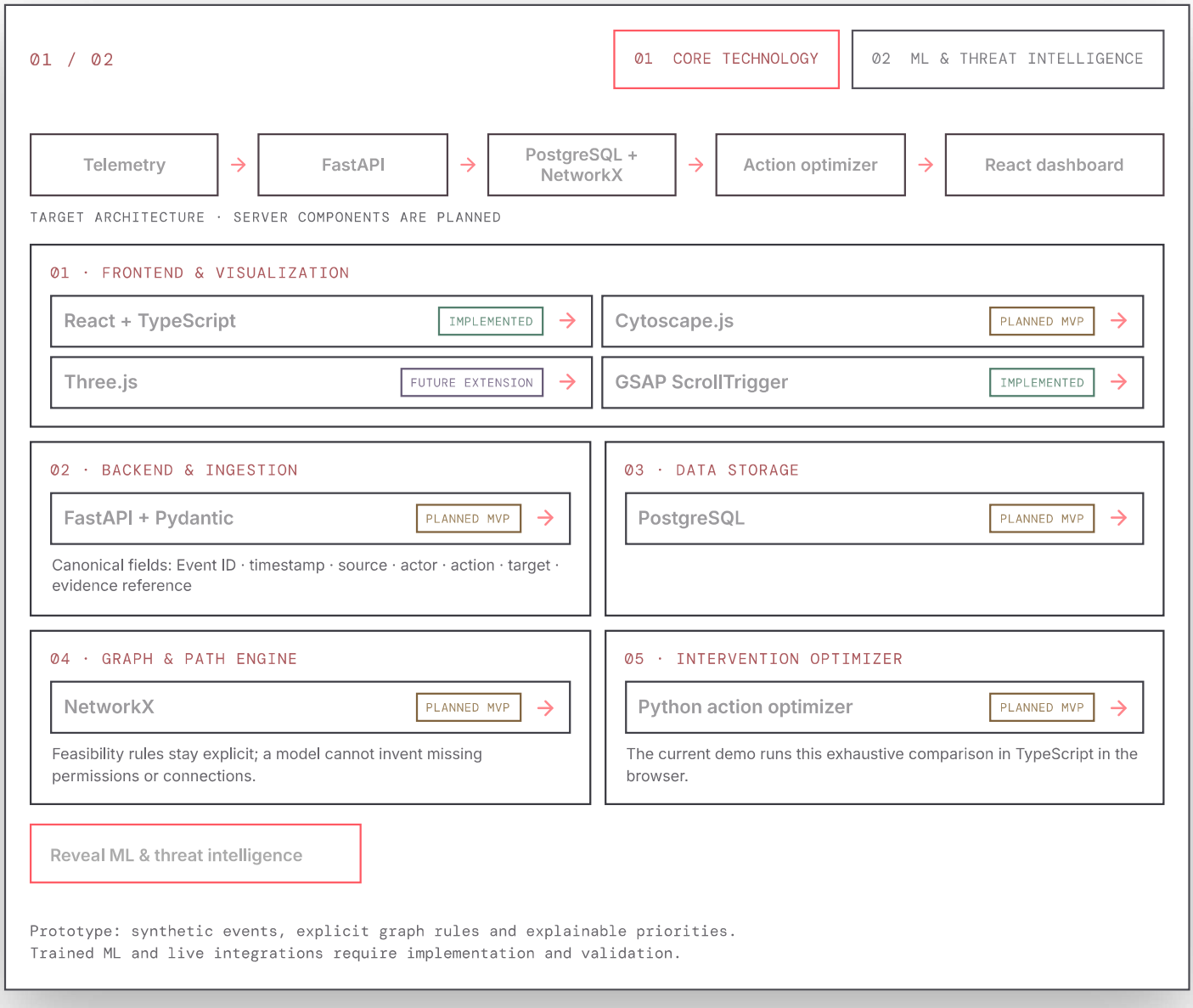
CONCLUSION AT EVENT 01

Key found in repository; its use is unconfirmed.

Next action: Confirm the key is active and mapped to this principal.

SCROLL STORY · EVENT FOLLOWS SCROLL

Inspect event evidence →


01 / 02
01 CORE TECHNOLOGY
02 ML & THREAT INTELLIGENCE
Telemetry
FastAPI
PostgreSQL + NetworkX
Action optimizer
React dashboard
TARGET ARCHITECTURE · SERVER COMPONENTS ARE PLANNED
01 · FRONTEND & VISUALIZATION
React + TypeScript
IMPLEMENTED
Cytoscape.js
PLANNED MVP
Three.js
FUTURE EXTENSION
GSAP ScrollTrigger
IMPLEMENTED
02 · BACKEND & INGESTION
FastAPI + Pydantic
PLANNED MVP
Canonical fields: Event ID · timestamp · source · actor · action · target · evidence reference
03 · DATA STORAGE
PostgreSQL
PLANNED MVP
04 · GRAPH & PATH ENGINE
NetworkX
PLANNED MVP
Feasibility rules stay explicit; a model cannot invent missing permissions or connections.
05 · INTERVENTION OPTIMIZER
Python action optimizer
PLANNED MVP
The current demo runs this exhaustive comparison in TypeScript in the browser.
Reveal ML & threat intelligence
Prototype: synthetic events, explicit graph rules and explainable priorities.
Trained ML and live integrations require implementation and validation.

SIMULATION ONLY · OBSERVED EVIDENCE IS PRESERVED

A	Exposed cloud credential POTENTIAL ROUTE	Critical
B	Vulnerable public application POTENTIAL ROUTE	High
C	Vendor integration exposure POTENTIAL ROUTE	Medium

SELECT A FIX

DISRUPTION POINTS

○ Revoke exposed cloud key	BREAKS A	1
○ Restrict public staging access	BREAKS B	2
○ Patch vulnerable staging service	BREAKS B	4
○ Rotate vendor integration token	BREAKS C	2

Apply lowest-cost A + B fixes

Reset

TOTAL 0 POINTS

Select fixes to block priority routes A and B.

Path C remains for separate review. Disruption points are illustrative, not financial costs.

10 | BUSINESS IMPACT · IBM RESEARCH

A breach is a business loss.

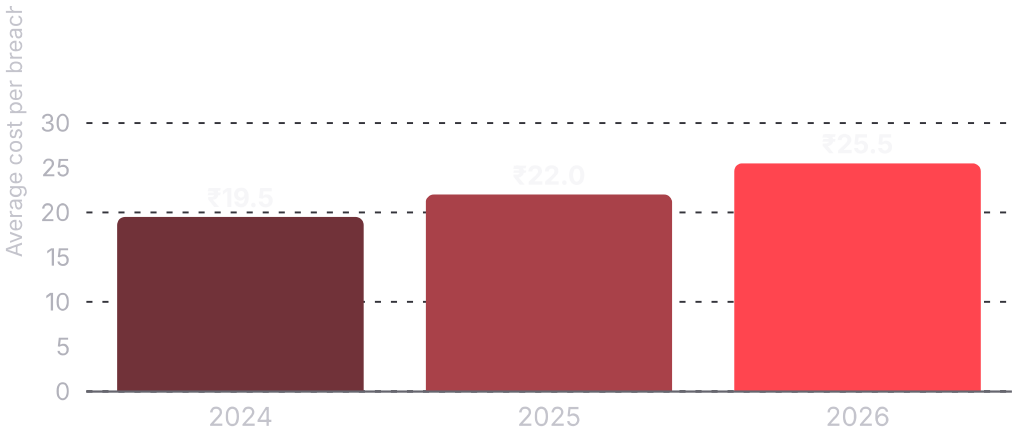
Published breach costs show why identifying dangerous routes matters.

AVERAGE ORGANIZATIONAL COST

Cost per breach

India | Global

Average cost per breach, ₹ crore



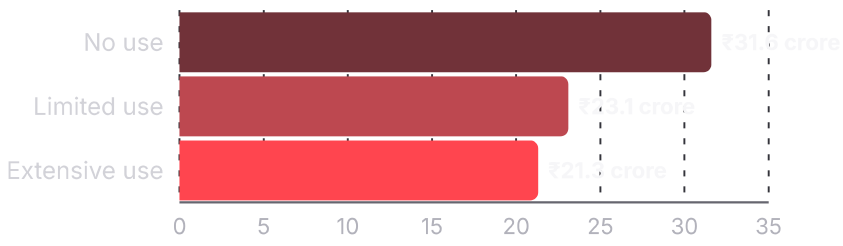
Hover, tap, or use the chart arrow keys. Exact values stay visible above the bars.

IBM India 2026 IBM India 2025 + 2024

INDIA · 2026

Breach cost by security AI and automation use

Average cost per breach · ₹ crore



An association reported across studied organizations.
Not guaranteed savings, causal proof or an AttackTwin benchmark.

IBM India 2026 source

Average organizational cost per breach in IBM's study. Report years are shown. Figures are not annual losses.

View data table

MANIPAL HACKATHON | EXPLORE THE ATTACKTWIN PROTOTYPE

One small signal.
A clearer next step.

Run the demo

View MITRE context ↓

ATTACKTWIN AI PREDICT. EXPLAIN. BREAK THE PATH.

BACK TO TOP ↑